

Dominion Samuel

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Education

Simon Fraser University

Bachelor of Science in Data Science | GPA: 4.15

Burnaby, BC

Expected Grad Date: December 2027

- **Courses:** Computational Data Science, Databases, Data Structures and Algorithms, Machine Learning, Artificial Intelligence
- **Awards & Leadership:** President's Honour Roll, Dean's Honour Roll, 2025 Data Science Award, Science Peer Mentor, ColorStack SFU Lead

Experience

Data Engineering Co-op

Samsung R&D Canada

Vancouver, BC

January 2026 – August 2026

- Migrated EKS-hosted Apache Airflow to AWS MWAA across production environments, delivering the release **3 weeks** ahead of schedule
- Developed a custom Cline skill to automate Airflow DAG syntax refactoring (v2.4 → v3.2) across **25+** repositories, removing manual rewrites
- Engineered GitHub Actions CI/CD workflows to package and deploy Airflow DAGs independently across isolated MWAA environments
- Eliminated cross-environment code drift and manual deployment steps by decoupling DAG releases for each MWAA environment
- Diagnosed Airflow metadata database bottlenecks through Argo worker pod inspection and Airflow DB shell analysis
- Purged orphaned Airflow metadata records identified during diagnosis, reducing system CPU usage from **90%** to **40%**
- Refactored PostgreSQL ETL architecture by shifting column-level transformations upstream, eliminating redundant downstream computation
- Accelerated AWS Redshift query processing time by **12%** through the upstream ETL redesign, lowering load on downstream tables
- Optimized Apache Spark jobs with broadcast joins and dynamic partition pruning to resolve data skew in daily pipelines
- Resolved daily pipeline bottlenecks behind production dashboards, eliminating recurring timeout failures

Projects

High-Throughput Ingestion Pipeline | [GitHub](#) | *Python, Polars, Docker, Astro Airflow, GCP*

August 2026

- Orchestrated a containerized Airflow pipeline with dynamic task mapping to parallelize **130,000+** daily API requests
- Spearheaded rate-limit handling through Airflow connection pools, eliminating **429** Too Many Requests failures at full volume
- Architected a Polars transformation layer to parse nested JSON payloads with strict schema validation into compressed Parquet
- Cut processing time by **40%** by replacing a prior Pandas-based approach with the Polars and Parquet transformation layer
- Pioneered in-memory streaming to Google Cloud Storage via io.BytesIO buffers, removing disk I/O and preventing corrupted uploads

Spotify Music Trends & Hit Prediction | [GitHub](#) | *Python, Pandas, NumPy, Scikit-learn, Jupyter Notebooks*

June 2025 – July 2025

- Engineered acoustic, lyrical, and temporal features from unstructured Spotify metadata spanning **20+ years** using Pandas and NumPy
- Trained K-Means and Random Forest models to classify songs by popularity, reaching **79%** accuracy on unseen tracks
- Uncovered the strongest predictors of popularity through feature importance analysis of the Random Forest model
- Published preprocessing steps, model pipelines, and evaluation metrics in a public GitHub repository to support reproducibility

E-Commerce Analytics Dashboard | [GitHub](#) | *SQL, MSSQL, Power BI, Azure*

January 2025 – April 2025

- Engineered MSSQL ETL queries using CTEs, window functions, and aggregates to transform raw transactional data into reporting-ready tables
- Accelerated query performance by **30%** through optimization of the ETL logic against unoptimized baseline queries
- Derived key e-commerce metrics, including customer churn rate, product profitability, and supplier performance, from order and inventory data
- Developed an interactive Power BI dashboard on Azure to surface retention and revenue trends across product, region, and time dimensions
- Empowered users to filter and explore **3 years** of historical operational data through the dashboard

Technical Skills

Languages: Python, SQL, C++

Data Engineering: ETL/ELT, Data Lakes, Data Warehousing, Apache Spark (PySpark, Spark SQL), Apache Airflow, Polars, Pandas, dbt, Parquet

Cloud & DevOps: AWS (S3, Glue, Lambda, Athena, MWAA, EKS, IAM), GCP (Cloud Storage), Docker, Terraform, Argo, GitHub Actions (CI/CD)

Databases (RDBMS & NoSQL): PostgreSQL, MSSQL, Amazon Redshift, DynamoDB

Reporting & Visualization: Power BI, Tableau, Excel

Developer Tools & AI Agents: Git/GitHub, Cline (AI Coding Agents, Prompt Engineering)